

Decision Making – Week 3

Project: Educational Microcontroller Simulator with Process Scheduling

Technology: Java **Target:** STC89C52

Week: 3 **Meeting Date:** 10-09-2026

1. Purpose of Decision Making

The Week 3 decisions were made to define the implementation approach for Memory, Stack, and FIFO Queue functionalities and to ensure that the new modules can be tested and integrated with the existing simulator.

2. Decisions Made

Decision	Description
Memory Implementation	Memory read and write operations will be implemented as part of the Week 3 memory functionality and connected with the existing simulator.
Stack Implementation	The Stack will use Stack Pointer management and support PUSH and POP operations. Stack Pointer updates will be verified during testing.
FIFO Queue	The Queue will follow First-In-First-Out (FIFO) behaviour using Enqueue and Dequeue operations.
Queue Flowchart	A flowchart will be prepared to clearly represent the steps involved in Queue operations.
Assembly Validation	An Assembly validation program will be prepared to verify the required Queue-related behaviour and execution.
Testing	Modules will be tested individually before integration, followed by overall testing after integration.
Documentation	Test cases, execution results, and relevant implementation details will be documented after testing and integration.

3. Reason for the Decisions

- The Memory and Stack components need to work with the existing simulator structure rather than being developed as separate standalone features.
- Stack Pointer tracking is required to maintain correct Stack behaviour during PUSH and POP operations.
- FIFO Queue behaviour must preserve the order in which elements are inserted and removed.
- Individual module testing will help identify errors before the modules are integrated into the complete simulator.
- Assembly validation and documented test results will provide evidence that the implemented functionality behaves as expected.

4. Responsibility-Based Decisions

Member	Decision / Responsibility
Neil Saldanha	Memory module implementation and integration with the existing simulator.
Shahala Shahzeen	Stack Pointer, PUSH/POP functionality, and Stack testing.
Thrupthi Anchan	FIFO Queue implementation, Enqueue/Dequeue testing, and Queue flowchart.
Aman Assadi	Assembly validation program, test cases, execution results, and documentation.

5. Expected Outcome

The decisions provide a clear implementation path for Week 3, with each module assigned to a specific team member and validated before final integration. The completed work will support continued development of the simulator and provide documented evidence of testing.